CASE STUDY | SUPPLY CHAIN AND LOGISTICS ENGINEERING | CHAMPIONSHIP

PortFlow Container Terminal: Yard and Gate Flow

Judges test whether you separate causes from symptoms, resist the obvious expansion, and use analytics and automation responsibly, covering finance, operations and safety, supply chain, change management and AI within $10M.

THE ANSWER

"Option B — Dynamic yard: demand zoning, MIP/CRAFT slotting, tracking database, AI AGV pilot — delivers 21.7% lower year-5 yard utilization (target 19.5%) for CAD $9,400,000 of the CAD $10,000,000 budget: NPV CAD $6,091,191 at 9.5%, payback 32.7 months. Five-year net financial impact (case formula): -CAD $16,100,000 (net savings)."

19.5%	$10.00M	12 mo	$6.09M
TARGET	**BUDGET CEILING**	**DEADLINE**	**NPV OF ANSWER**
Lower year-5 yard utilization	capital, CAD	core implementation	at 9.5% WACC

INSIDE

01 The Challenge
02 The Approach
03 The Options
03b The Package
04 The Results
05 Execution
05b Required Analysis
06 Defence & Submission

Archetype: Operations & engineering improvement · Family: Throughput / capacity · Model: Baseline KPIs → bottleneck/root cause → optimize/simulate → pilot → ROI + KPIs

SITUATION · COMPLICATION · QUESTION

What this case is really asking

Judges test whether you separate causes from symptoms, resist the obvious expansion, and use analytics and automation responsibly, covering finance, operations and safety, supply chain, change management and AI within $10M.

S SITUATION

PortFlow is a Canadian container terminal linking shipping lines, trucking, rail and inland customers. Today the yard holds 21,120 TEU against 24,000 TEU effective capacity (88%), containers dwell 6 days, and trucks take 72 minutes per visit (40 queuing, 32 inside). Volume will grow 20% in five years. The terminal spends $6M a year on overtime and $8M on delay and service penalties. Release and schedule information is not shared consistently, planning runs across separate systems on fixed rules, and the yard system knows a container's stack but not its exact position. One 15-acre parcel (about 4,000 TEU) could be added, phased around live operations.

C COMPLICATION

With +20% volume the yard passes capacity by year 5, trucks already take 72 minutes against a 45-minute objective, overtime ($6M) and penalties ($8M) keep rising, and the one expansion parcel would still leave the yard about 90% full.

Q QUESTION

Which package, within the $10M upfront budget, attacks PortFlow's core issue so that at +20% volume the yard runs at or below 85% utilization and trucks turn in under 45 minutes by year five?

A ANSWER

Option B — Dynamic yard: demand zoning, MIP/CRAFT slotting, tracking database, AI AGV pilot — delivers 21.7% lower year-5 yard utilization (target 19.5%) for CAD $9,400,000 of the CAD $10,000,000 budget: NPV CAD $6,091,191 at 9.5%, payback 32.7 months. Five-year net financial impact (case formula): -CAD $16,100,000 (net savings).

CONSTRAINTS

- Upfront investment at most $10M CAD; it does not need to be spent i...
- Year-5 objectives: average yard utilization <= 85% and average truc...
- Vessel and truck operations cannot stop while changes are introduced
- Customs holds and missing documentation cannot be overridden by Por...
- Up to 8 content slides for a 10-minute presentation

WHAT THE JUDGES SCORE



SOLUTION MODEL · OPERATIONS & ENGINEERING IMPROVEMENT

How we cracked it

Baseline KPIs → bottleneck/root cause → optimize/simulate → pilot → ROI + KPIs

How does PortFlow reach <= 85% yard utilization and < 45-minute truck turns at +20% volume, within $10M?

Placement (where boxes sit)

- Which container groups move most, and how far are they from their exit?
- Can MIP + CRAFT cut flow x distance?

Order and timing (when they leave)

- Do stacks follow pickup order?
- How does seasonal demand shift, and can slotting follow it?

Information and movement

- Do we know each box's exact slot and release status?
- Can AGVs move boxes cheaper and safer than manual tractors?

HYPOTHESIS (FORMED IN THE FIRST 20 MINUTES)

"PortFlow's yard is static and blind. Demand-based zoning (MIP + CRAFT), dynamic slotting on one tracked database, and shared release status cut handling time and dwell enough to meet both objectives without new land; AI AGVs are piloted, not bet on."

TOOLKIT

- MIP: container-group-to-block assignment
- CRAFT: pairwise exchange layout improvement
- Demand forecasting with seasonality
- Decision matrix with budget as a hard gate

QUICK MATH

- Today: 21,120 / 24,000 TEU = 88% full
- Year 5 at today's stay: about 25,300 TEU, over capacity
- Expansion only: about 90% (misses 85%)
- Truck turn: 72 min today; plan 15 queue + 22 inside = 37 min
- Budget: $9.4M of $10M; five-year impact -$16.1M

TIME PLAN FOR A 3-HOUR CRACKING WINDOW

0:00-0:25	0:25-0:55	0:55-1:40	1:40-2:30	2:30-3:00
Read & align	Quick math	Package & finance	Deck (8 slides)	Rehearse & Q&A
SCQA, cause-vs-symptom tree, first hypothesis	Case KPIs, budget check, truck turn gap	Initiatives, costs, owners, 5-year formula	Answer first, matrix, budget check, plan, risks	10-minute run, likely judge questions

SOLUTION LADDER

Four levels of solution, one recommendation

Option B is recommended. It is both the cheapest compliant option and the best value: a dominant choice.

LEVEL 1 · QUICK WIN

Option A

3.9% · $1.60M

Lowest capital: what a minimal move buys

LEVEL 2 · MINIMUM COMPLIANT

Option B

21.7% · $9.40M

Cheapest option that meets the target within budget

RECOMMENDED

LEVEL 3 · VALUE MAXIMISER

Option B

21.7% · $9.40M

Highest NPV within budget

RECOMMENDED

LEVEL 4 · STRETCH

Option B

21.7% · $9.40M

Largest improvement regardless of budget

RECOMMENDED

DECISION MATRIX (SCORE 0-10, WEIGHTED)

Criterion	Weight	Option A	Option B (chosen)	Option C
Target attainment	30%	1.3	7.4	4.9
Financial return (NPV)	25%	9.2	10.0	2.0
Payback speed	15%	6.7	4.6	0.0
Downside resilience (-15% perf.)	15%	1.7	9.5	6.2
Budget headroom	15%	8.4	0.6	0.0
Weighted total	100%	5.2	6.9	2.9

THE THREE ALTERNATIVES

Option A: Static re-zoning only: one-off MIP and CRAFT layout

$1.60M · 3.9% · NPV $4.03M
In budget · misses target

Option B: Dynamic yard: demand zoning, MIP/CRAFT slotting, tracking database, AI AGV pilot

$9.40M · 21.7% · NPV $6.09M
In budget · meets target

Option C: Yard expansion on the 15-acre parcel with two added RTGs

$14.50M · 14.3% · NPV -$14.63M
OVER BUDGET · misses target

RECOMMENDED PACKAGE

Dynamic yard: demand zoning, MIP/CRAFT slotting, tracking database, AI AGV pilot

Every initiative has an upfront cost, an ongoing cost and a named owner for its savings.

INITIATIVE & PURPOSE	UPFRONT	PER YEAR	OWNER
Yard zoning and block layout (MIP + CRAFT) MIP assigns container groups to blocks by move frequency and exit; CRAFT swaps zones while flow x distance falls; re-marking and phased moves.	$1.20M	$100K	Yard Planning Manager
Dynamic slotting engine with demand forecasting Re-slots containers as demand shifts (e.g. a December peak for one product) and stacks by planned pickup; planners approve.	$1.60M	$300K	Yard Planning Manager
Tracking and yard database (digital twin) OCR cameras, RFID tags on vehicles and crane position sensors log every move into one yard database.	$2.20M	$350K	IT and Operations Manager
Release visibility and truck pickup windows The database shares release and hold status with lines, truckers, rail and customers; trucks book windows for ready boxes.	$1.40M	$250K	Commercial Director
AI AGV pilot: 2 units in a fenced lane Two battery AGVs, fleet software linked to the database, navigation, charging and safety fencing; scale only if KPIs pass.	$1.60M	$150K	Engineering and Maintenance Manager
Change management and paid training Co-design workshops, super-users per shift, AGV supervisor and technician training in paid time.	$600K	$150K	HR Director with the Employee Representative
Contingency (about 9%) Released at the phase-3 gate if unused.	$800K	$0	Finance Director

BUDGET CHECK

$9.40M committed (94%) **$600K headroom · budget $10.00M**

FIVE-YEAR NET FINANCIAL IMPACT (CASE FORMULA)

$9.40M + 5 x ($1.30M - $6.40M) = -$16.10M

Investment + 5 x (additional annual costs - annual savings); undiscounted; negative = net savings. Discounted cross-check: NPV $6.09M at 9.5%, payback 32.7 mo.

DIRECTIONS CONSIDERED · SAME FORMULA

Option A: Static re-zoning only: one-off MIP and CRAFT layout
Set aside: a one-off layout goes stale as demand shifts and does not cut dwell, so year-5 utilization stays above 100%.
-$7.40M
upfront $1.60M

Option B: Dynamic yard: demand zoning, MIP/CRAFT slotting, tracking database, AI AGV pilot
Recommended.
-$16.10M
upfront $9.40M

Option C: Yard expansion on the 15-acre parcel with two added RTGs
Set aside: costs about $14.5M (over budget) and still misses the objective at 90.5% utilization; it treats the symptom.
$13.50M
upfront $14.50M · over budget

ECONOMICS · PROFIT-DRIVEN

What the recommendation is worth

Run-rate benefit $6.40M/yr net of $1.30M/yr O&M; levelized 134474.76 CAD per year per % point of improvement.

$6.09M	28.1%	32.7 mo	1.42x
NET PRESENT VALUE	**IRR**	**PAYBACK**	**BENEFIT / COST**
5 yrs @ 9.5%	annual	monthly cash model	present value

CASH FLOW BY YEAR (BARS) AND CUMULATIVE (LINE)



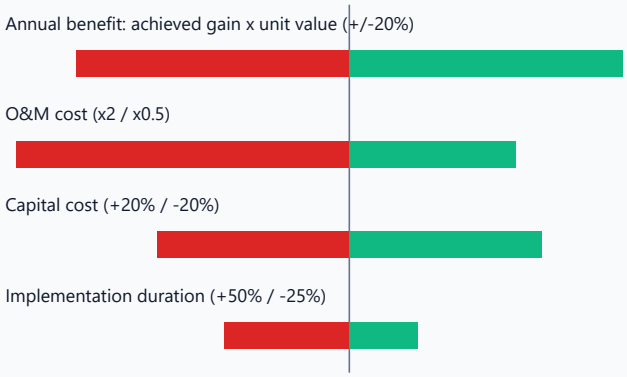
SCENARIOS

Bear (-15% perf., +20% capex)
18.45% · NPV $140K · payback 46.7 mo · misses target

Base (approved plan)
21.7% · NPV $6.09M · payback 32.7 mo · meets target

Bull (+15% perf., -10% capex)
24.95% · NPV $10.60M · payback 26.4 mo · meets target

NPV SENSITIVITY (TORNADO)



30 / 60 / 90 DAYS

From pilot to control

One owner per workstream; every phase ends in a go/no-go gate.

DAYS 1–30

Validate & pilot

- Confirm baseline data and the binding constraint
- Launch pilot of Option B at the constraint
- Name owners; start KPI dashboard

DAYS 31–60

Prove & decide

- Pilot gate: Layout model beats today's travel by >= 20% in simulation; AGV safety case signed
- Lock fixed-price packages for scale-up
- Train operators; write standard work

DAYS 61–90

Scale & control

- Begin re-zone, go dynamic, pilot 2 agvs
- Control chart on the primary KPI
- Review vs target (19.5%) with sponsor

IMPLEMENTATION HORIZONS

Phase 1 · Months 1-3 **Design: data, layout model, AGV safety case**	KPI: Move data cleaned; MIP/CRAFT layout validated in simulation; AGV vendor and safety case approved Gate: Layout model beats today's travel by >= 20% in simulation; AGV safety case signed	**$2.20M**
Phase 2 · Months 4-9 **Re-zone, go dynamic, pilot 2 AGVs**	KPI: Dwell <= 5.3 days; turn <= 55 min; rehandles -20%; AGVs commissioned Gate: Slotting advice accepted >= 70%; AGV availability >= 90% in the fenced lane	**$5.60M**
Phase 3 · Months 10-12 **Scale decision and hand over**	KPI: Dwell <= 4.8 days; turn < 45 min; utilization on track for <= 85% Gate: AGV cost per move <= manual tractor before any fleet expansion; savings confirmed	**$1.60M**

TOP RISKS

AGV breakdown or navigation fault (Medium / High)

Mitigation: Preventive maintenance plan and vendor SLA, sensor health monitoring, fenced lane, manual tractors as instant fallback
Trigger: AGV availability < 90% or any unplanned stop in the lane

Tracking or database errors (wrong slot recorded) (Medium / High)

Mitigation: OCR and crane-sensor double check on every move, nightly reconciliation audit, manual search procedure kept
Trigger: Position accuracy < 99% in the daily audit

AI slotting or demand forecast wrong (Medium / Medium)

Mitigation: Re-slots run weekly on actual bookings; planners approve; segregation rules locked; 6 weeks of shadow mode
Trigger: Override rate > 30% or rehandles rise for 2 weeks

KPIS: BASELINE → TARGET

Average yard dwell time

6.0 days → <= 4.8 days (plan 4.7)

Year-5 yard utilization at +20% volume

88% today; 106% if nothing changes → <= 85% (plan about 83%)

Truck turn time

72 min (40 queue + 32 inside) → < 45 min (plan about 37)

Rehandles per container

measured in phase 1 → -30%

AGV pilot availability and cost per move

manual tractors → >= 90% uptime; cost per move <= manual

Overtime and penalties

$6M and $8M a year → $3.6M and $4.0M a year

WHAT THE BRIEF ASKS FOR

Required analysis, area by area

Operations and safety, supply chain, change management, and AI and automation, each answered from the case data.

OPERATIONS & SAFETY

• Storage: shared release status and pickup windows cut dwell from 6.0 to 4.7 days, keeping year-5 utilization near 83% without new land.
• Container movement: MIP/CRAFT zoning puts high-demand groups next to their exit; dynamic slotting stacks by pickup order, so cranes stop digging.
• Truck service: queue 40 to about 15 min with pickup windows; time inside 32 to about 22 min with fast movers near the gate: about 37 minutes in total.
• Safety: AGVs run only in a fenced lane; segregation rules for hazardous, reefer and weight are locked in the model; every go-live has a safety review.
• Measures, reviewed weekly: utilization, dwell, rehandles per box, travel per move, queue and inside time, AGV uptime, incidents and near-misses.

SUPPLY CHAIN

• Shipping lines: vessel ETA changes flow into the database and trigger re-slotting before the ship arrives.
• Truckers: pickup windows only for released, ready containers, already staged near the gate.
• Rail: rail-bound containers zoned next to the rail exit and staged by train departure.
• Customers: release, hold and ETA alerts; seasonal demand signals (for example a December peak) feed the slotting forecast.
• Customs holds stay outside PortFlow's control but are flagged early so no truck is booked against a held box.

ORGANIZATIONAL CHANGE MANAGEMENT

• Stakeholders: yard operators, tractor drivers, planners, maintenance, IT, truckers, rail, lines, customers and finance.
• Involve staff before tools arrive: planners and operators co-design zones and slotting rules in phase 1.
• Train in paid time: super-users per shift; tractor drivers can certify as AGV supervisors or technicians.
• Clarify roles: no layoffs from this program; a RACI for every zone, model and AGV decision.
• Keep operations running: one block re-zoned at a time, overnight moves, AGVs only in a fenced lane, rollback plan.

AI & AUTOMATION

• MIP: assigns container groups (customer, product, direction) to blocks to minimise move frequency x distance to their exit, within block capacity and segregation rules.
• CRAFT: tests pairwise swaps of zones and keeps any swap that lowers total flow x distance, so the layout can be improved again as demand changes.
• Dynamic slotting: a demand forecast (seasonal peaks) plus actual bookings re-slot containers weekly; planners approve each change.
• AI AGVs: 2 battery units in a fenced quay-to-block lane, tracked by the yard database; overseen by the Engineering and Maintenance Manager with preventive maintenance.
• Performance checks: 6 weeks of shadow mode for slotting; AGV uptime, cost per move and incidents weekly; scale the fleet only if the month-12 gate passes.

BEFORE YOU SUBMIT

Judge questions and the final checklist

14 prepared answers, all computed from the model. Answer first, then evidence.

STRESS TEST

Why not just build on the 15-acre parcel?

Space alone fails the objective: 25,344 / 28,000 TEU = 90.5% at year 5, for about $14.5M, over budget. Cutting dwell to 4.7 days reaches about 83% on today's footprint.

STRESS TEST

Why only two AGVs?

The evidence is mixed: successful automated terminals cut operating costs 25-55%, but some lose 7-15% productivity. Two units in a fenced lane prove uptime, cost per move and safety for $1.6M; the fleet grows only if the month-12 gate passes.

STRESS TEST

How do the MIP and CRAFT work together?

The MIP assigns container groups to blocks, minimising move frequency x distance to their exit within capacity and segregation rules. CRAFT then tries pairwise zone swaps and keeps any that lower total flow x distance, so the layout can be re-improved as demand shifts.

STRESS TEST

What if the seasonal forecast is wrong?

Slotting re-runs weekly on actual bookings, not only the forecast; planners approve every re-slot; zoning is reviewed monthly.

STRESS TEST

How do you know where every container is?

OCR cameras read container numbers at the gate and quay, crane position sensors record the exact slot of every pick and drop, and RFID tags identify each AGV and truck. Every move writes to one yard database that the slotting engine, the AGVs and the customer portal all read.

STRESS TEST

Will automation cost jobs, and who maintains the AGVs?

No layoffs from this program. Drivers can retrain as AGV supervisors and technicians. AGVs follow a preventive maintenance plan with a vendor SLA, and manual tractors stay as the fallback.

PRE-SUBMISSION CHECKLIST

✓ Answer-first executive summary (one sentence, three numbers)	Option B — Dynamic yard: demand zoning, MIP/CRAFT slott…
✓ Three options costed on the same basis	decision matrix with weights shown
✓ Recommendation fits the budget	CAD $9,400,000 of CAD $10,000,000
✓ Recommendation meets the target	21.7% vs 19.5%
✓ Financials: NPV, payback, sensitivity	NPV CAD $6,091,191, payback 32.7 months
✓ Bear case tested (-15% performance, +20% cost)	bear NPV CAD $140,445
✓ 30/60/90 roadmap with owners and gates	pilot → prove → scale
✓ Top risks with triggers	6 risks logged
✓ Q&A bank of 10+ prepared answers	see judge_QA_preparation.md
! Assumptions validated with real data	replace: Monetisation, Discount rate (WACC), Appraisal horizon, Annual O&M, Benefit during i…
✓ Safety & licensed engineering review noted	MOC + HAZOP before commissioning

WHERE THIS CASE STYLE APPEARS (DIRECTORY-LISTED; WINNERS NOT VERIFIED)

- Boeing Northwest Case Competition · USA
- Colorado State University Supply Chain Case Competition · USA
- Carlson National Undergraduate Supply Chain Case Competition · USA
- University of St. Thomas Supply Chain Case Competition · USA
- Kohler Undergraduate Wisconsin Supply Chain Cup · USA